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Effective conductivity of periodic arrays of spheres with interfacial resistance

BY H. CHENG AND S. TORQUATO

Department of Civil Engineering and Operations Research and Princeton Materials Institute, Princeton University, Princeton, NJ 08544, USA

We consider the problem of analytically determining the effective thermal conductivity of a composite material consisting of periodic arrays of spheres with interfacial resistance. We applied Rayleigh's method which has been used extensively for such calculations in the perfect interface case, i.e. no jump in the temperature across the interface. Results are presented for simple, body-centred and face-centred cubic arrays, and each for a wide range of volume fractions. Our calculations were based on very accurate lattice sums in all three lattice cases and temperature fields were resolved to very high multipole moments, including all azimuthal terms. In the case of zero interfacial strength, our formulation recovers all previously reported benchmark results.

1. Introduction

There exists a vast literature on predicting the effective conductivity of two-phase composite materials (Rayleigh 1892; Batchelor 1974; Willis 1981; Hashin 1983; Torquato 1991). Virtually all of these studies adopt the assumption of a perfect-interface between the constituents, implying that the potential field and the normal component of the flux are continuous across the two-phase interface. In a variety of applications, the perfect interface assumption cannot be made. For example, thermal resistance at the interface (due to roughness or acoustic mismatch between the phases) can dramatically decrease the effective conductivity of the composite (Chiew & Glandt 1983; Benveniste 1987; Every *et al.* 1992; Lipton & Vernescu 1996). This interfacial resistance is measured by a jump in the temperature across the interface. Recently, Torquato & Rintoul (1995) used classical minimum energy principles to find very sharp, rigorous bounds on the effective properties of a class of composites with imperfect interfaces that incorporates crucial microstructural information about the interface.

In this paper, we have generalized an analytical tool—Rayleigh's method—to find the effective conductivity of periodic arrays of spherical inclusions with imperfect interfaces. Despite the fact that it was devised more than 100 years ago, Rayleigh's method has received a great deal of attention in recent years due to the dramatic increase in computational power. It has been applied by a variety of investigators (McPhedran & McKenzie 1978; McKenzie *et al.* 1978; Sangani & Acrivos 1983; Hinson & Felderhof 1992; Greengard & Moura 1994) to study the perfect-interface case.

We extend Rayleigh's method in the present work to take into account the effect of the interface in terms of a dimensionless parameter R which characterizes the inter-

facial resistance. Our approach is essentially an extension of the work of McKenzie *et al.* (1978), but we take into account all of the azimuthal terms, and we choose a different unit cell in the body-centred and face-centred cubic lattice cases which results in different lattice sums and different linear systems to be solved numerically. Another feature of our calculation is that we have used extremely accurate lattice sums (Berman & Greengard 1995), and the solution is resolved until it exhibits a spectral decaying rate in non-singular cases. Hence our results achieved spectral accuracy in describing both the field and the effective conductivity.

The remainder of the paper is organized as follows: in §2, we describe in detail the boundary-value problem for the temperature field and the solution techniques. In §3, an exact formula for effective conductivity has been derived to give results to the order of $O(f^9)$, where f is the volume fraction of the spheres. In §4, we present all of our numerical results and discuss them. In §5, we show that an approximation expression for the effective conductivity that incorporates a three-point microstructural parameter (see Torquato & Rintoul 1996) predicts the effective conductivity remarkably well.

2. Formulation and solution technique

(a) Formulation of the problem

We consider a composite material consisting of periodic arrays of equal-sized spherical inclusions of radius a and conductivity σ_2 embedded in a uniform matrix of conductivity σ_1 , in which there exists a thermal resistance at the interface between the two phases. The strength of this resistance is characterized by a dimensionless parameter R which can be defined through a limiting process by first examining a three-phase composite in which each inclusion possesses a concentric coating of thickness δ and conductivity σ_c and then letting $\delta \rightarrow 0$ and $\sigma_c \rightarrow 0$ (see Torquato & Rintoul 1996), i.e.

$$R = \frac{\tilde{R}\sigma_2}{a}, \quad \text{with} \quad \tilde{R} = \lim_{\substack{\delta \rightarrow 0 \\ \sigma_c \rightarrow 0}} \frac{\delta}{\sigma_c}. \quad (2.1)$$

Unlike the perfect interface case, this is equivalent to prescribing a temperature jump across the interface, namely,

$$J_n = \frac{1}{R}[T_- - T_+]. \quad (2.2)$$

where T_+ and T_- are the temperatures as one approaches the inclusion surface from the matrix and particle side respectively, and

$$J_n = -\sigma_1 \frac{\partial T_+}{\partial n} = -\sigma_2 \frac{\partial T_-}{\partial n} \quad (2.3)$$

is the normal component of the heat flux at the interface which is continuous across the interface.

Without loss of generality, let us apply a uniform field with strength E_0 along the negative z -axis to the composite material. The harmonic temperature field is governed by the following boundary value problem inside the cubic unit periodic cell Ω with sides of length 1:

$$\nabla \cdot (\sigma \nabla T) = 0 \quad \text{in} \quad \Omega, \quad (2.4)$$

$$\sigma(\mathbf{x}) = \begin{cases} \sigma_2 & \text{if } \mathbf{x} \in V, \\ \sigma_1 & \text{if } \mathbf{x} \in \Omega \setminus V, \end{cases}$$

$$\frac{\sigma_2}{Ra}(T_+ - T_-) = \sigma_2 \frac{\partial T_-}{\partial n} = \sigma_1 \frac{\partial T_+}{\partial n} \quad \text{on } \partial V, \quad (2.5)$$

with periodic boundary conditions given by

$$\begin{aligned} T(x+1, y, z) &= T(x, y, z), \\ T(x, y+1, z) &= T(x, y, z), \\ T(x, y, z+1) &= T(x, y, z) - E_0. \end{aligned} \quad (2.6)$$

Here V is the volume occupied by the spherical inclusion, ∂V is the sphere surface, and $\partial/\partial n$ is the outward normal derivative on ∂V .

In the case of a perfect interface ($R = 0$), a formal solution to the above partial differential equation can be obtained analytically using the spherical-harmonics-expansion procedure devised by Lord Rayleigh (1891) in his pioneering work. For our case ($R > 0$), we will show in the next subsection that the similar procedure can be carried out to obtain an analytical solution with its coefficients satisfying an infinite-dimensional linear system of equations.

After solving equations (2.4)–(2.6), our remaining task is to find the effective conductivity for the composite material. In the perfect interface case, Lord Rayleigh and many others showed that a simple application of Green's theorem could accomplish this, yielding a formula for the effective conductivity in terms of the dipole moment of the solution. For the imperfect interface case, this becomes much more complicated because the definition of proper average quantities in the composite is different from that in a composite with perfect interfaces, which will now includes a surface integral involving the temperature jump across the interface, and it is impossible to rely on Green's theorem again in our case. But we will show in §2e that the effective conductivity can be derived again from the temperature field by using the field representations directly and surprisingly we arrived at the same formula as in the perfect interface case.

(b) Solution technique

To solve equations (2.4)–(2.6), we represent the temperature field in terms of spherical harmonics (Hobson 1955; Jackson 1975; Wallace 1985; Greengard 1988). The potential inside the sphere is given by

$$T_2(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n C_{nm} r^n Y_n^m(\theta, \phi), \quad (2.7)$$

and the potential outside the sphere by

$$T_1(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n \left(A_{nm} r^n + \frac{B_{nm}}{r^{n+1}} \right) Y_n^m(\theta, \phi), \quad (2.8)$$

where functions of the form $Y_n^m(\theta, \phi)r^n$ are the regular solid harmonics of degree n , while those of form $Y_n^m(\theta, \phi)/r^{n+1}$ are called irregular solid harmonics of degree $-n-1$. Here (r, θ, ϕ) is the spherical coordinate of point $\mathbf{x}$. Following the notation of Greengard (1988), the Y_n^m are defined in terms of the associated Legendre functions

by

$$Y_n^m(\theta, \phi) = \sqrt{\frac{(n-|m|)!}{(n+|m|)!}} \cdot P_n^{|m|}(\cos \theta) e^{im\phi}. \quad (2.9)$$

With these representations, equation (2.4) is automatically satisfied. Imposition of the boundary condition (2.5) on the sphere surface and the use of orthogonal properties of spherical harmonics yields

$$\begin{aligned} \sigma_2 C_{nm} n a^{n-1} &= \sigma_1 \left(A_{nm} n a^{n-1} - (n+1) \frac{B_{nm}}{a^{n+2}} \right) \\ &= \frac{\sigma_2}{Ra} \left[(A_{nm} - C_{nm}) a^n + \frac{B_{nm}}{a^{n+1}} \right], \end{aligned}$$

from which we obtain

$$A_{nm} = \frac{B_{nm}}{a^{2n+1}} \cdot \frac{\alpha + (n+1)/n + (n+1)R}{1 - \alpha + nR}, \quad (2.10)$$

$$C_{nm} = \frac{B_{nm}}{a^{2n+1}} \cdot \frac{2n+1}{n(1 - \alpha + nR)}, \quad (2.11)$$

where $\alpha = \sigma_2/\sigma_1$ is the conductivity ratio. However, the relations above are not enough to uniquely determine the coefficients A_{nm} , B_{nm} and C_{nm} . These coefficients are determined by requiring that our solution representations satisfy a consistency condition, which also enforces the periodic boundary conditions (2.6):

$$\sum_{n=0}^{\infty} \sum_{m=-n}^n A_{nm} r^n Y_n^m(\theta, \phi) = E_0 z + \sum_{k \neq 0} \sum_{n=0}^{\infty} \sum_{m=-n}^n \frac{B_{nm}}{\rho_k^{n+1}} Y_n^m(\theta_k, \phi_k), \quad (2.12)$$

where the index k runs over all spheres except the one at the origin, and $(\rho_k, \theta_k, \phi_k)$ is the spherical coordinate of the point (r, θ, ϕ) when measured in the local coordinate system centred at the k^{th} sphere. The last term can be expanded in terms of coordinates (r, θ, ϕ) by using the addition theorem (Greengard 1988) for spherical harmonics:

$$\frac{Y_n^m(\theta_k, \phi_k)}{\rho_k^{n+1}} = \sum_{n=0}^{\infty} \sum_{m=-n}^n G(n, m; i, j; k, 0) r^n Y_n^m(\theta, \phi), \quad (2.13)$$

where

$$G(n, m; i, j; k, 0) = (-1)^n J_j^m \cdot \frac{H_i^j H_n^m}{H_{n+i}^{j-m}} \cdot \frac{Y_{n+i}^{m-j}(\alpha_{k0}, \beta_{k0})}{\rho_{k0}^{n+i+1}},$$

and $(\rho_{k0}, \alpha_{k0}, \beta_{k0})$ is the spherical coordinate of the k th sphere centre in the current spherical coordinate system, and $H_n^m, J_m^{m'}$ are defined by

$$H_n^m = \frac{(-1)^n}{\sqrt{(n-m)! \cdot (n+m)!}}, \quad (2.14)$$

$$J_m^{m'} = \begin{cases} (-1)^{\min(|m'|, |m|)}, & \text{if } m \cdot m' > 0; \\ 1, & \text{otherwise.} \end{cases} \quad (2.15)$$

Substituting (2.13) and (2.10) to (2.12), using the orthogonal properties of the spherical harmonics once again, we arrive at (after some rearrangements) the following

infinite linear system of equations for B_{nm} :

$$\frac{B_{nm}}{a^{2n+1}} \frac{\alpha + (n+1)/n + (n+1)R}{1 - \alpha + nR} = E_0 \delta_{n1} \delta_{m0} + \sum_{i=0}^{\infty} \sum_{j=-i}^i (-1)^i J_m^j \mathcal{L}_{n+i}^{j-m} \Upsilon_{n+m}^{i-j} \Upsilon_{n-m}^{i+j} B_{ij} \quad (2.16)$$

for $n = 0, 1, \dots, \infty$, $m = -n, \dots, n$. Here δ_{ij} is the standard Kronecker delta,

$$\mathcal{L}_n^m = \sum_{p \in \Lambda_3} Y_n^m(\theta_p, \phi_p) / r_p^{n+1}, \quad n \geq 1, \quad m = -n, \dots, 0, \dots, n \quad (2.17)$$

are three-dimensional lattice sums, where Λ_3 denotes the set

$$\{(k_1, k_2, k_3) \mid k_i \in \mathbf{Z}, (k_1, k_2, k_3) \neq (0, 0, 0)\}.$$

And Υ_k^l represents

$$\Upsilon_k^l = \sqrt{\binom{k+l}{k}} \quad (2.18)$$

for any integer k, l . The linear system (2.16) is valid in all three cubic lattice cases that we are going to deal with. The difference among them lies at the fact that (2.17) will have different values for each lattice case.

From (2.16), we can see that the effect of the interface is to change its left-hand side. When $R = 0$, (2.16) reduces to the linear systems of references McPhedran & McKenzie (1978) and McKenzie *et al.* (1978).

(c) Numerical solutions

In practice, the infinite system (2.16) has to be made finite by ignoring all multipole moments beyond a given order. In addition, the full symmetry of cubic arrays makes it possible to greatly reduce the number of unknowns of (2.16). First we observe that, for our choice of the unit cell described in the next subsection, the temperature is an odd function of θ , and hence only those spherical harmonics with odd degree in n are needed. Moreover, it must be a symmetric function of ϕ about $\phi = \pi/4$, thus only those spherical harmonics whose order m is an exact multiple of 4 can contribute. After truncation and some simplification, (2.16) becomes

$$\begin{aligned} & \frac{B_{2n-1,m}}{a^{4n-1}} \frac{\alpha + 2n/(2n-1) + 2nR}{1 - \alpha + (2n-1)R} \\ & + \sum_{i=1}^K \sum_{j=0}^{2i-1} B_{2i-1,j} \frac{\epsilon_j}{2} [\mathcal{L}_{2(n+i-1)}^{j-m} \Upsilon_{2n-1+m}^{2i-1-j} \Upsilon_{2n-1-m}^{2i-1+j} + \mathcal{L}_{2(n+i-1)}^{j+m} \Upsilon_{2n-1-m}^{2i-1-j} \Upsilon_{2n-1+m}^{2i-1+j}] \\ & = E_0 \delta_{n1} \delta_{m0}, \end{aligned} \quad (2.19)$$

for $n = 1, 2, \dots, K$, $m = 0, 4, \dots, m \leq 2n-1$. Here m, j run over only numbers which are exact multiples of 4, and ϵ_j is the Neumann symbol,

$$\epsilon_j = \begin{cases} 1, & \text{if } j = 0; \\ 2, & \text{otherwise.} \end{cases}$$

Notice also that the linear system (2.19) is symmetric. Relation (2.19) is the final linear system which we solve numerically for simple, body-centred and face-centred cubic lattice of spherical inclusions.

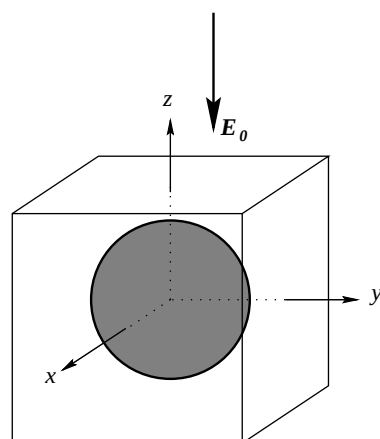


Figure 1. The simple cubic lattice structure.

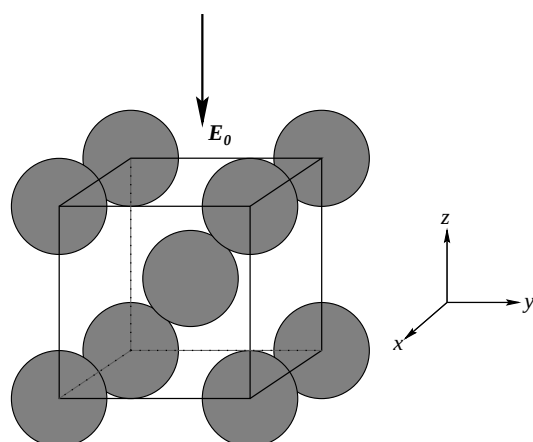


Figure 2. The body-centred cubic lattice structure.

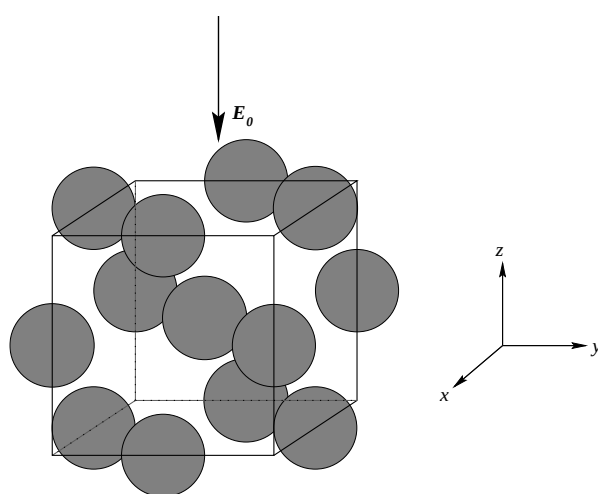


Figure 3. The face-centred cubic lattice structure.

(d) Choice of unit cell and lattice sums

Before we present our numerical results, we need to clarify our choices of the unit computational cell and the corresponding lattice sums. Certainly, one can not change material properties by choosing different computational cells, but the choice can affect the computing process.

Figure 1, 2 and 3 show the unit cells and the location of the spherical inclusions used in our calculations. They are all conventional, but differ from the approach taken in McPhedran & McKenzie (1978) and McKenzie *et al.* (1978) for body and face-centred cubic lattice cases. The lattice sums for the simple cubic lattice have been computed very accurately by Berman & Greengard (1995) using the renormalization method. We have generalized this method in order to generate the lattice sums for body and face-centred cubic lattice cases. The non-zero lattice sums for $n \leq 20$ and for all three lattice cases are listed in the table in the Appendix.

(e) Effective conductivity

The question we want to address now is that of obtaining the effective conductivity from the solution of (2.16). Extra attention is needed here for our composite materials whose constituent interfaces exhibit contact resistance, since the definition of the average intensity, $\langle \mathbf{E} \rangle$, and the average flux, $\langle \mathbf{J} \rangle$, are now given by

$$\langle \mathbf{E} \rangle = -\frac{1}{V} \int_V \nabla T \, dV + \frac{1}{V} \int_{\partial S} (T_1 - T_2) \mathbf{n} \, dS, \quad (2.20)$$

$$\langle \mathbf{J} \rangle = \frac{1}{V} \int_V \mathbf{J} \, dV. \quad (2.21)$$

Notice that the average potential gradient, $\langle \mathbf{E} \rangle$, in (2.20) contains an additional integral involving the temperature jump across the interface ∂S . Of course, if the temperature field is continuous across the interface (perfect interface case), then (2.20) reduces to the well-known classical form.

The effective thermal conductivity tensor of the material is now defined via the relation,

$$\langle \mathbf{J} \rangle = \boldsymbol{\sigma}_{\text{eff}} \langle \mathbf{E} \rangle, \quad (2.22)$$

where $\boldsymbol{\sigma}_{\text{eff}}$ is generally a tensor, but for our isotropic material, it is characterized by a single scalar parameter σ_{eff} through $\boldsymbol{\sigma}_{\text{eff}} = \sigma_{\text{eff}} \mathbf{I}$, where $\mathbf{I}$ is the 3×3 unit matrix. We now show that the quantity σ_{eff} can be determined from the solution of (2.19) by the same formula as in McPhedran & McKenzie (1978) and McKenzie *et al.* (1978), even though the definition of $\langle \mathbf{E} \rangle$ has been changed to include the imperfect interface. We will use a different approach by applying the representation of T directly.

Consider the simple cubic lattice case for which $V = 1$, thus

$$\begin{aligned} \langle \mathbf{J} \rangle &= \int_{V_1} \mathbf{J} \, dV + \int_{V_2} \mathbf{J} \, dV \\ &= \sigma_1 \int_{V_1} \nabla T_1 \, dV + \sigma_2 \int_{V_2} \nabla T_2 \, dV \\ &= \sigma_1 \left[\langle \mathbf{E} \rangle - (1 - \alpha) \int_{V_2} \nabla T_2 \, dV - \int_{\partial S} (T_1 - T_2) \mathbf{n} \, dS \right], \end{aligned}$$

using the fact that $\nabla^2 T_2 = 0$ and the boundary condition (2.5), the above equation

Table 1. Numerical constants $c_1, \dots, c_6$

	simple cubic	body-centred cubic	face-centred cubic
c_1	1.30472	0.12930	0.07529
c_2	0.07232	0.25696	0.24195
c_3	-0.52895	-0.09881	0.05583
c_4	0.15256	0.01313	0.02311
c_5^*	-0.30667	0.05338	-0.05244
c_6	0.01045	0.00562	9.16×10^{-7}

Table 2. Dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ of simple cubic lattice of spherical inclusions with imperfect interfaces versus the volume fraction f (Here $f_{\text{max}} = \pi/6$. The critical value $R_c = 9$ for $\alpha = 10$ and $R_c = 9999$ for $\alpha = 10000$.)

f	conductivity ratio $\alpha = 10$		conductivity ratio $\alpha = 10000$	
	$R = 5$	$R = 30$	$R = 5000$	$R = 20000$
0.00	1.00000	1.00000	1.00000	1.00000
0.05	1.02752	0.95688	1.03796	0.97030
0.10	1.05555	0.91497	1.07690	0.94117
0.15	1.08410	0.87420	1.11683	0.91259
0.20	1.11318	0.83447	1.15780	0.88453
0.25	1.14279	0.79567	1.19982	0.85694
0.30	1.17292	0.75766	1.24289	0.82975
0.35	1.20356	0.72027	1.28702	0.80290
0.40	1.23470	0.68329	1.33218	0.77628
0.425	1.25045	0.66487	1.35514	0.76303
0.45	1.26630	0.64643	1.37833	0.74978
0.475	1.28225	0.62790	1.40174	0.73651
0.50	1.29829	0.60913	1.42535	0.72316
0.51	1.30473	0.60151	1.43485	0.71778
0.515	1.30795	0.59767	1.43961	0.71507
0.52	1.31118	0.59379	1.44437	0.71235
0.523	1.31311	0.59145	1.44723	0.71072

can be rewritten as

$$\langle \mathbf{J} \rangle = \sigma_1 \left[\langle \mathbf{E} \rangle - (1 - \alpha) \int_{\partial S} \mathbf{x} \frac{\partial T_2}{\partial n} dS - Ra \int_{\partial S} \mathbf{n} \frac{\partial T_2}{\partial n} dS \right],$$

where $\mathbf{x}$ is the vector (x, y, z) . Now substitute (2.7) into the above equation and note that $\langle \mathbf{E} \rangle = (0, 0, E_0)$, after some computation, one obtains for the simple cubic lattice case,

$$\langle \mathbf{J} \rangle = \sigma_1 (1 - 4\pi B_{1,0}/E_0) \langle \mathbf{E} \rangle, \quad (2.23)$$

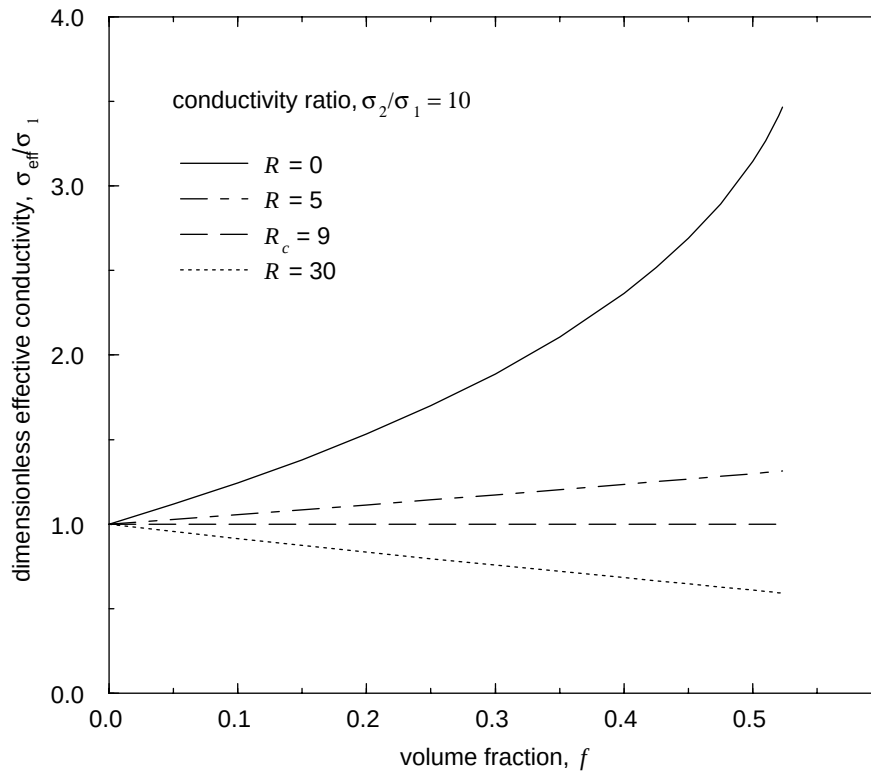


Figure 4. Dimensionless effective conductivity $\sigma_{\text{eff}}/\sigma_1$ versus volume fraction f for a simple cubic lattice with sphere to matrix conductivity ratio $\alpha = \sigma_2/\sigma_1 = 10$. The solid line is the previously known result for the perfect-interface case ($R = 0$). The dash-dotted line is for a moderate interfacial resistance ($R = 5$), the long-dashed line is for the critical interfacial resistance value ($R_c = 9$), and the dotted line for a large interfacial resistance ($R = 30$).

from which one can easily obtain the formula to compute σ_{eff}

$$\sigma_{\text{eff}} = \sigma_1(1 - 4\pi B_{1,0}/E_0). \quad (2.24)$$

Similarly, one can show that for the body-centred cubic lattice,

$$\sigma_{\text{eff}} = \sigma_1(1 - 8\pi B_{1,0}/E_0), \quad (2.25)$$

and for the face-centred cubic lattice,

$$\sigma_{\text{eff}} = \sigma_1(1 - 16\pi B_{1,0}/E_0). \quad (2.26)$$

Despite the modified definition (2.20) for the average intensity, relations (2.24)–(2.26) have exactly the same forms as in the perfect interface case, but here the coefficient $B_{1,0}$ incorporates the effect of the imperfect interface.

3. The exact solution for σ_{eff} to $O(f^9)$

In this section, we shall solve the exact solution for σ_{eff} from (2.19) to all multipoles of order 2^9 , including all the azimuthal terms. We obtained

$$\frac{\sigma_{\text{eff}}}{\sigma_1} = 1 - \frac{3f}{D}, \quad (3.1)$$

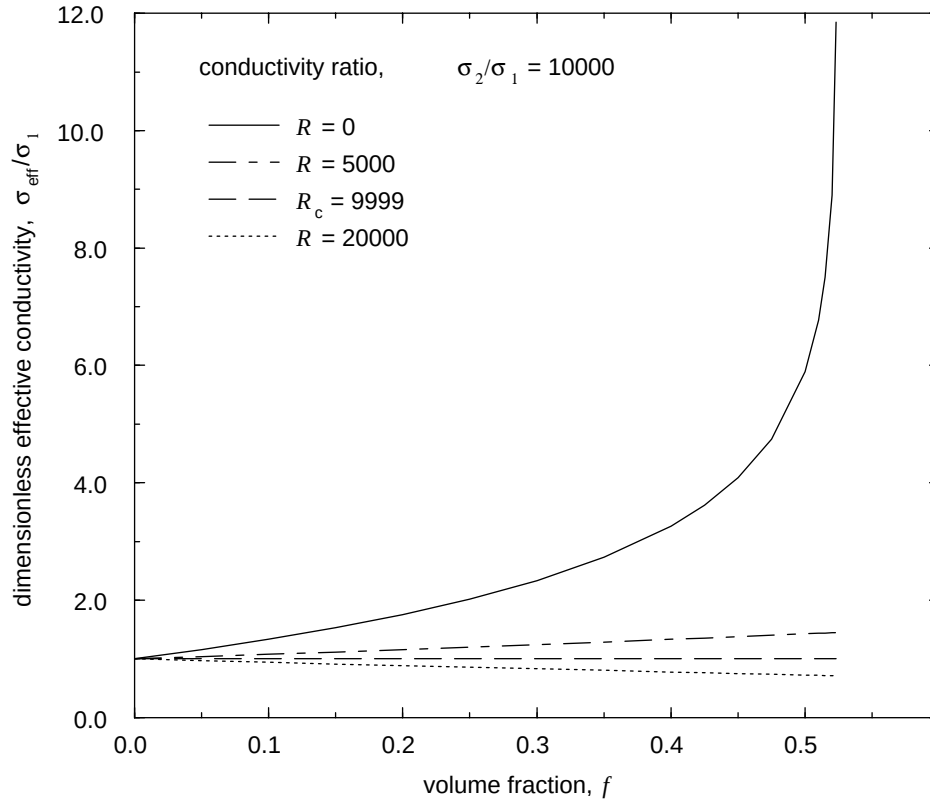


Figure 5. Dimensionless effective conductivity $\sigma_{\text{eff}}/\sigma_1$ versus volume fraction f for a simple cubic lattice with sphere to matrix conductivity ratio $\alpha = \sigma_2/\sigma_1 = 10000$. The solid line is the previously known result for the perfect-interface case ($R = 0$). The dash-dotted line is for a moderate interfacial resistance ($R = 5000$), the long-dashed line is for the critical interfacial resistance value ($R_c = 9999$), and the dotted line for a strong interfacial resistance ($R = 20000$).

where

$$D = -\beta_1^{-1} + f + c_1\beta_3f^{10/3} + c_2\beta_5f^{14/3} + c_3\beta_3^2f^{17/3} \\ + c_4\beta_7f^6 + c_5\beta_3\beta_5f^7 + c_6\beta_9f^{22/3} + O(f^{25/3}),$$

in which, β_i is defined by

$$\beta_i = \frac{\alpha - 1 - iR}{\alpha + (i+1)/i + (i+1)R}$$

and numerical constants $c_1, \dots, c_6$ for three cubic lattice arrays are computed and listed in table 1.

Formula (3.1) enables one to estimate the effective conductivity to the order of $O(f^9)$ for composite materials with imperfect interfaces, with the interfacial resistance incorporated in the constants β_i . Our calculated results of constants $c_1, \dots, c_6$ agreed extremely well with the results reported in Sangani & Acrivos (1982) for the perfect interface case, except a sign error occurred at c_5 in Sangani & Acrivos (1982). The results in McPhedran & McKenzie (1978) and McKenzie *et al.* (1978) were in-

Table 3. *Dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ of body-centred cubic lattice of spherical inclusions with imperfect interfaces versus the volume fraction f*
 (Here $f_{\text{max}} = \sqrt{3}\pi/8$. The critical value $R_c = 9$ for $\alpha = 10$ and $R_c = 9999$ for $\alpha = 10000$.)

f	conductivity ratio $\alpha = 10$		conductivity ratio $\alpha = 10000$	
	$R = 5$	$R = 30$	$R = 5000$	$R = 20000$
0.00	1.00000	1.00000	1.00000	1.00000
0.05	1.02752	0.95688	1.03796	0.97030
0.10	1.05556	0.91498	1.07690	0.94117
0.15	1.08411	0.87424	1.11684	0.91261
0.20	1.11320	0.83462	1.15784	0.88460
0.25	1.14284	0.79604	1.19991	0.85710
0.30	1.17304	0.75845	1.24310	0.83010
0.35	1.20380	0.72176	1.28742	0.80356
0.40	1.23513	0.68589	1.33291	0.77745
0.425	1.25100	0.66822	1.35608	0.76454
0.45	1.26701	0.65072	1.37955	0.75171
0.475	1.28315	0.63337	1.40330	0.73897
0.50	1.29943	0.61615	1.42732	0.72630
0.525	1.31583	0.59903	1.45163	0.71368
0.55	1.33236	0.58200	1.47621	0.70113
0.575	1.34901	0.56501	1.50104	0.68860
0.60	1.36576	0.54803	1.52611	0.67609
0.625	1.38261	0.53099	1.55140	0.66356
0.65	1.39954	0.51378	1.57688	0.65098
0.66	1.40633	0.50682	1.58712	0.64591
0.67	1.41312	0.49978	1.59737	0.64082
0.68	1.41993	0.49263	1.60764	0.63568

complete due to the fact that all the azimuthal terms were neglected, as pointed out by Sangani & Acrivos (1982).

4. Results and discussion

The general method that we have used here is well-established and has been proven to give accurate results in the perfect interface case (McPhedran & McKenzie 1978; McKenzie *et al.* 1978). We have checked our algorithm with the results reported there and found excellent agreement in general. We will not repeat the perfect interface results here but instead will focus on the imperfect interface case. We implemented the method in double precision in FORTRAN with the finite linear system (2.19) being solved directly by LAPACK subroutine DSYSV, which is an indefinite symmetric linear system solver, a well suited tool for our purpose.

The results for simple cubic lattice arrays are presented in table 2 and figures 4 and 5. Columns 2 and 3 in table 2 list the dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ with conductivity ratio $\alpha = \sigma_2/\sigma_1 = 10$ for two different values of the

Table 4. *Dimensionless effective thermal conductivity $\sigma_{\text{eff}}/\sigma_1$ of face-centred cubic lattice of spherical inclusions with imperfect interfaces versus the volume fraction f*
(Here $f_{\text{max}} = \sqrt{3}\pi/8$. The critical value $R_c = 9$ for $\alpha = 10$ and $R_c = 9999$ for $\alpha = 10000$.)

f	conductivity ratio $\alpha = 10$		conductivity ratio $\alpha = 10000$	
	$R = 5$	$R = 30$	$R = 5000$	$R = 20000$
0.00	1.00000	1.00000	1.00000	1.00000
0.05	1.02752	0.95688	1.03796	0.97030
0.10	1.05556	0.91498	1.07690	0.94117
0.15	1.08411	0.87425	1.11684	0.91261
0.20	1.11320	0.83462	1.15784	0.88460
0.25	1.14285	0.79606	1.19991	0.85711
0.30	1.17305	0.75849	1.24311	0.83012
0.35	1.20382	0.72183	1.28744	0.80359
0.40	1.23515	0.68600	1.33295	0.77750
0.425	1.25103	0.66837	1.35614	0.76461
0.45	1.26705	0.65091	1.37963	0.75180
0.475	1.28321	0.63360	1.40340	0.73908
0.50	1.29950	0.61642	1.42746	0.72643
0.525	1.31593	0.59936	1.45181	0.71385
0.55	1.33247	0.58240	1.47643	0.70132
0.575	1.34915	0.56550	1.50131	0.68884
0.60	1.36594	0.54864	1.52646	0.67639
0.625	1.38284	0.53179	1.55184	0.66395
0.65	1.39983	0.51489	1.57744	0.65150
0.675	1.41691	0.49789	1.60324	0.63902
0.70	1.43406	0.48071	1.62922	0.62647
0.71	1.44093	0.47375	1.63965	0.62142
0.72	1.44781	0.46674	1.65009	0.61634
0.73	1.45469	0.45964	1.66066	0.61123
0.74	1.46158	0.45244	1.67101	0.60608

dimensionless resistance R and a wide range of volume fractions. Figure 4 shows corresponding plots of these results. Columns 4 and 5 of table 2 and figure 5 are for arrays with an virtually infinite contrast ratio, i.e. $\alpha = 10000$. The figures also include results for the perfect interface case (solid lines).

It can be seen that the interfacial resistance has a dramatic effect on the effective property. It has been shown that there exists a critical value $R_c = \alpha - 1$ of the dimensionless parameter R such that when $R = R_c$ the effective conductivity of the composite material equals exactly the matrix conductivity (Torquato & Rintoul 1995), i.e. the inclusions are effectively ‘hidden’. More significantly, when $R > R_c$, the more conducting spherical inclusions actually reduce the conductivity of the composite material below that of the matrix. This effect can be seen from the figure where the long dashed lines represent the results when R equals the critical value R_c . The critical value R_c is directly related to the notion of a critical radius first

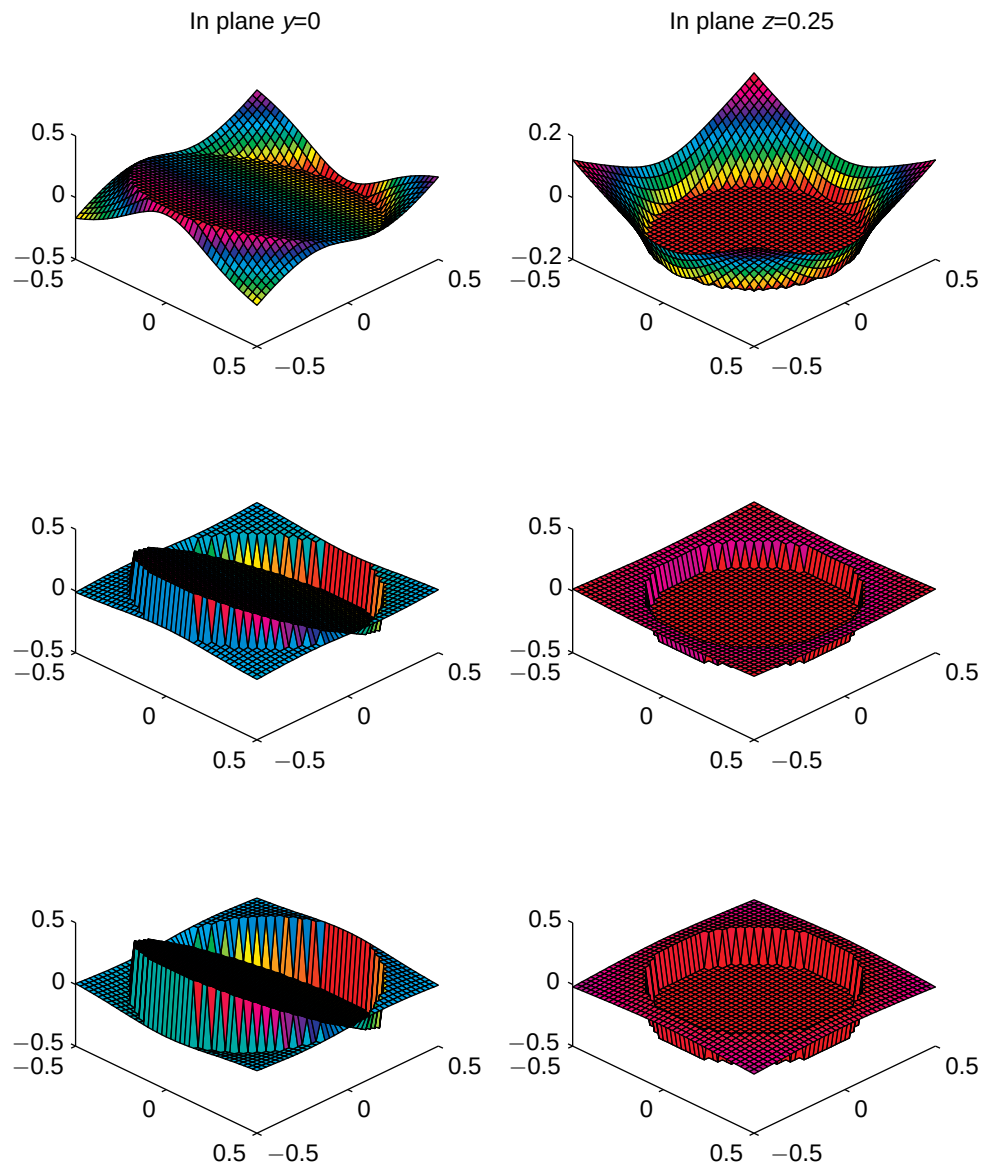


Figure 6. The surface plot of potential in plane $y = 0$ (first column) and plane $z = 0.25$ (second column). The material is composed of a simple cubic lattice of spheres with conductivity ratio $\alpha = 10$ and volume fraction $f = 0.5$. The first row is for $R = 0$ (no interfacial resistance), the second is for $R = 5$ and the third for $R = 30$. The jump of the potential at the interface is shown clearly in the latter two cases.

observed by others (Chiew & Glandt 1983; Every *et al.* 1991; Lipton & Vernescu 1996).

Tables 3 and 4 give corresponding results for body-centred and face-centred cubic arrays, respectively. Qualitatively, these results are similar to those for the simple cubic array illustrated in figures 4 and 5 and hence are not depicted graphically.

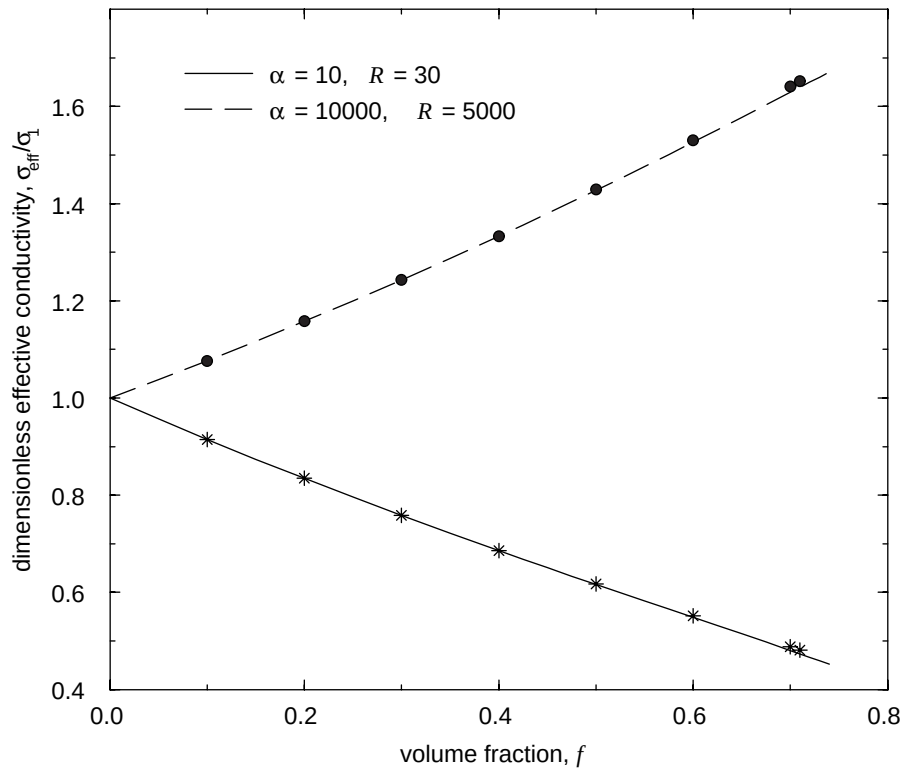


Figure 7. Dimensionless effective conductivity $\sigma_{\text{eff}}/\sigma_1$ versus volume fraction f for face-centred cubic lattices of spheres. The solid line is the exact numerical result from §4 for conductivity ratio $\alpha = 10$ and interfacial resistance $R = 30$, while the dashed line is for $\alpha = 10000$ and $R = 5000$. The stars and filled circles represent the respective predicted values of formula (5.1).

Finally, we present in figure 6 surface plots of the temperature fields in the plane $y = 0$ and $z = 0.25$ for the simple cubic lattice case for the conductivity ratio $\alpha = 10$ and volume fraction $f = 0.5$. The first, second and third rows of Figure 6 corresponds to $R = 0$, $R = 5$ and $R = 30$ respectively. The results for body-centred and face-centred cubic arrays are similar and hence are not plotted.

5. An accurate approximation formula

The asymptotic formula (3.1) we presented in §3 can be used to estimate the effective conductivity, but it yields accurate results only for moderate values of the volume fraction. For example, in the simple cubic lattice case, we examined and found that (3.1) provides useful results only when $f < 0.3$ for the imperfect interface case. In this section, we will present a simpler but much more powerful formula (5.1), based not just on the volume fraction alone, but also on a three-point microstructural parameter ζ_2 . In the perfect interface case, such a formula has been shown (Torquato 1985) to be able to predict the effective conductivity remarkably well for both periodic and random arrays of inclusions and for a wide range of volume fractions and conductivity ratios. We will show here that an appropriately generalized

Table 5. The non-zero lattice sums $\mathcal{L}_n^m$ for $n \leq 20$

n	m	simple cubic	body-centred cubic	face-centred cubic
2	0	$2\pi/3$	$4\pi/3$	$8\pi/3$
4	0	0.310822E+01	-0.310646E+01	-0.752569E+01
4	4	0.185752E+01	-0.185647E+01	-0.449746E+01
6	0	0.573329E+00	0.544656E+01	-0.266349E+02
6	4	-0.107260E+01	-0.101896E+02	0.498293E+02
8	0	0.325929E+01	0.764839E+01	0.811865E+02
8	4	0.122566E+01	0.287618E+01	0.305302E+02
8	8	0.186744E+01	0.438222E+01	0.465166E+02
10	0	0.100922E+01	-0.939666E+01	-0.152435E+01
10	4	-0.101696E+01	0.946866E+01	0.153603E+01
10	8	-0.121042E+01	0.112700E+02	0.182824E+01
12	0	0.289125E+01	0.486715E+01	-0.938077E+02
12	4	0.891680E+00	-0.692780E+01	-0.281749E+03
12	8	0.106997E+01	0.139589E+02	0.329947E+03
12	12	0.158878E+01	-0.244497E+00	-0.139104E+03
14	0	0.115364E+01	0.152759E+02	-0.291581E+03
14	4	-0.848340E+00	-0.112333E+02	0.214418E+03
14	8	-0.910342E+00	-0.120543E+02	0.230089E+03
14	12	-0.110537E+01	-0.146368E+02	0.279381E+03
16	0	0.279236E+01	-0.120685E+02	0.909996E+03
16	4	0.800719E+00	0.197893E+02	0.406632E+03
16	8	0.841132E+00	-0.156788E+02	0.198649E+03
16	12	0.947940E+00	-0.277459E+02	0.160735E+03
16	16	0.150058E+01	0.284767E+01	0.547506E+03
18	0	0.125802E+01	-0.847180E+01	-0.391631E+02
18	4	-0.749851E+00	0.146809E+01	0.492731E+03
18	8	-0.784928E+00	0.206436E+02	-0.198831E+04
18	12	-0.845512E+00	-0.117252E+02	0.230920E+04
18	16	-0.104366E+01	0.123007E+02	-0.658506E+03
20	0	0.270422E+01	0.346397E+02	-0.121337E+04
20	4	0.709961E+00	-0.309409E+02	-0.350522E+04
20	8	0.734552E+00	0.162299E+02	0.213314E+03
20	12	0.783313E+00	0.358415E+02	0.170274E+04
20	16	0.891188E+00	0.433054E+02	0.213845E+04
20	20	0.141533E+01	0.408147E+00	-0.204563E+04

version of the formula for $R > 0$ (Torquato & Rintoul 1996) can also provide very good estimate of the effective conductivity in the imperfect interface case.

Torquato & Rintoul (1996) derived the following approximate formula for the effective conductivity σ_{eff} of a suspension (random or periodic) with the dimensionless

resistance R ,

$$\frac{\sigma_{\text{eff}}}{\sigma_1} = \frac{1 + 2f\beta_1 - 2(1-f)\zeta_2\beta_1^2}{1 - f\beta_1 - 2(1-f)\zeta_2\beta_1^2}, \quad (5.1)$$

where the parameter β_1 is given in §3, and ζ_2 is the aforementioned three-point microstructural parameter. In the perfect interface case, β_1 reduces to the parameter β_{21} contained in the approximate formula of Torquato (1985). For the three cubic lattices of spheres, ζ_2 has been tabulated as a function of the volume fraction f by McPhedran & Milton (1981).

Figure 7 compares the exact numerical results for $\sigma_{\text{eff}}/\sigma_1$ from §4 to corresponding results predicted by formula (5.1) for the face-centred cubic lattice case. It is seen that formula (5.1) provides an excellent estimate of $\sigma_{\text{eff}}/\sigma_1$ for different values of conductivity ratio, interfacial resistance and a wide range of volume fractions. The predictions of (5.1) for the simple and body-centred cubic lattice cases are equally good and hence are not shown graphically. In the perfect-interface case (Torquato 1985), (5.1) was shown to give good estimates for random configurations as well. Torquato & Rintoul (1996) have shown that (5.1) yields good estimates of the effective conductivity of random dispersions with imperfect interfaces.

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Appendix A. The lattice sums

We list in table 5 the non-zero lattice sums for $n \leq 20$ and for all three cubic lattice arrays. For details of the renormalization method that we used to calculate these sums and more complete results for the simple cubic lattice sums, we refer to the original paper by Berman & Greengard (1995).

References

- Batchelor, G. K. 1974 Transport properties of two-phase materials with random structure. *A. Rev. Fluid Mech.* **6**, 227–255.
- Benveniste, Y. 1987 Effective thermal conductivity of composites with a thermal contact resistance between the constituents: non-dilute case. *J. Appl. Phys.* **61**, 2840–2843.
- Berman, C. L. & Greengard, L. 1994 A renormalization method for the evaluation of lattice sums. *J. Math. Phys.* **35**, 6036–6048.
- Bonnecaze, R. T. & Brady, J. F. 1990 A method for determine the effective conductivity of dispersions of particles. *Proc. R. Soc. Lond. A* **430**, 285–313.
- Chiew, Y. C. & Glandt, E. D. 1983 The effect of structure on the conductivity of a dispersion. *J. Coll. Interface Sci.* **94**, 90–104.
- Every, A. G., Tzou, Y., Hasselman, D. P. H. & Raj, R. 1992 The effect of particle size on the thermal conductivity of ZnS/diamond composites. *Acta Metall. Mater.* **40**, 123–129.
- Greengard, L. 1988 *The rapid evaluation of potential fields in particle systems*. Cambridge, MA: MIT Press.
- Greengard, L. & Moura, M. 1994 On the numerical evaluation of the electrostatic fields in composite materials. *Acta Numerica*, pp. 379–410. Cambridge University Press.
- Hashin, Z. 1983 Analysis of composite materials. *J. Appl. Mech.* **50**, 481–504.
- Hinsen, K. & Felderhof, B. U. 1992 Dielectric constant of a suspension of uniform spheres. *Phys. Rev. B* **46**, 12955–12963.
- Hobson, E. W. 1955 *The theory of spherical and ellipsoidal harmonics*. New York: Chelsea.
- Proc. R. Soc. Lond. A* (1997)

- Jackson, J. D. 1975 *Classical electrodynamics*. New York: Wiley.
- Lipton, R. & Vernescu, B. 1996 Composites with imperfect interfaces. *Proc. R. Soc. Lond. A* **452**, 329–358.
- McPhedran, R. C. & Milton, G. W. 1981 Bounds and exact theories for the transport properties of inhomogeneous media. *Appl. Phys. A* **26**, 207–220.
- McPhedran, R. C. & McKenzie, D. R. 1978 The conductivity of lattices of spheres. I. The simple cubic lattice. *Proc. R. Soc. Lond. A* **359**, 45–63.
- McKenzie, D., McPhedran, R. & Derrick, G. 1978 The conductivity of lattices of spheres. II. The body centred and face centred cubic lattices. *Proc. R. Soc. Lond. A* **362**, 211–232.
- McPhedran, R. C., McKenzie, D. R. & Phan-Thien, N. 1983 Transport properties of two-phase composite materials. In *Advances in the mechanics and the flow of granular materials*, pp. 413–482. Clausthal: Trans. Tech. Publications.
- Rayleigh, Lord 1892 On the influence of obstacles arranged in rectangular order upon the properties of a medium. *Phil. Mag.* **34**, 481–502.
- Sangani, A. S. & Acrivos, A. 1982 The effective conductivity of a periodic array of spheres. *Proc. R. Soc. Lond. A* **386**, 263–275.
- Torquato, S. 1985 Effective electrical conductivity of two-phase disordered composite media. *J. Appl. Phys.* **58**, 3790–2797.
- Torquato, S. 1991 Random heterogeneous media: microstructure and improved bounds on effective properties. *Appl. Mech. Rev.* **44**, 37–76.
- Torquato, S. & Lado, F. 1988 Bounds on the conductivity of a random array of cylinders. *Proc. R. Soc. Lond. A* **438**, 591–604.
- Torquato, S. & Rintoul, M. 1995 Effects of the interface on the properties of composite media. *Phys. Rev. Lett.* **52**, 2635–2638.
- Torquato, S. & Rintoul, M. 1996 Conductivities of suspensions with imperfect interfaces. (In preparation.)
- Van Bladel, J. 1964 *Electromagnetic fields*. New York: McGraw-Hill.
- Wallace P. R. 1985 *Mathematical analysis of physical problems*. New York: Dover.
- Willis, J. R. 1981 Variational and related methods for the overall properties of composite materials. In *Advances in applied mechanics* (ed. C.-S. Yih), vol. 21, pp. 2–78. New York: Academic Press.

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